SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	M M Amal
Student ID	JF/ICT/24/15
Implementation Status	Completed
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need to download student records for administrative reporting, record-keeping, and external processing. Before CR-015, student records could be viewed within the system, but there was no dedicated facility to export them as a CSV file.

1.2 Existing Behaviour

Before CR-015 was implemented, administrators could view student records in the Student Management System. However, the system did not provide an option to download those records as a CSV file. Administrators who needed the information for reporting or record-keeping had to view or copy it manually.

The existing student management features, including adding, viewing, editing, and deleting student records, were available through the administrator interface. Access to student records was protected by the system's existing authentication and administrator authorization controls.

1.3 Requested Change

CR-015 introduces a CSV export facility to the administrator student management page. It allows an authorized administrator to download available student records in a portable format without manually copying the information.

2. Implementation

2.1 Implementation Summary

The **Export CSV** feature was added to the administrator student management page. It uses the student records already retrieved and displayed by the dashboard to generate a downloadable CSV file.

The file contains the following columns:

- Student ID
- Registration Number
- First Name
- Last Name
- Email
- Phone Number

The CSV includes a header row and is downloaded as students.csv. Values are escaped to preserve CSV formatting when they contain quotation marks, commas, or line breaks. A UTF-8 byte order mark is included to improve compatibility with spreadsheet applications.

The export reuses the existing student records API. No separate export endpoint, database modification, or change to the authentication mechanism was required.

2.2 Affected Files

File	Change	Purpose
frontend/src/pages/DashboardPage.jsx	Modified	Added CSV generation, the export handler, and the Export CSV control.
backend/src/routes/studentRoutes.js	No change	Existing authentication and administrator authorization protect the student-list API used by the export.
Database schema	No change	The feature uses the existing student data structure.

2.3 Technical Changes

- Added an **Export CSV** control to the administrator student management page.
- Generated the CSV from the student records already loaded by the dashboard.
- Included the required CSV column headers and student fields.
- Escaped values for correct CSV formatting.
- Set the downloaded filename to students.csv.
- Reused the existing student-list API, which requires authentication and the administrator role.
- Kept the export operation read-only; it does not create, update, or delete student records.

2.4 Implementation Logic

- The administrator opens the student management page.
- The dashboard retrieves student records through the existing protected student-list API.
- When the administrator clicks **Export CSV**, the system selects the required fields from the loaded records: Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- The system adds the column headers and formats each value for CSV, escaping quotation marks and other special characters.
- The generated CSV is converted into a downloadable file named students.csv.
- The browser downloads the file to the administrator's device. The operation does not modify the student records.

2.5 Error and Data Handling

- The export uses the student records already loaded by the dashboard. If the records cannot be retrieved, the existing API error handling applies.
- If the student list is empty, the CSV can still be generated with the required headers and no data rows.
- CSV values are escaped to preserve commas, quotation marks, and line breaks in student data.
- A UTF-8 byte order mark is included to improve compatibility with spreadsheet applications.
- Access to the records remains protected by the existing authentication and administrator authorization controls.
- □ Exporting is read-only and does not add, update, or delete student data.

3. Testing & Verification

3.1 Testing Approach

Field	Details
Testing Types	Functional testing, CSV content verification, authorization testing, and regression testing.
Environment	Development environment
Testing Method	Manual testing
Testing Status	Completed

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the administrator student management page	Export CSV control is available to the administrator.	Control displayed successfully.	Pass
TC-02	Export student records	A CSV file downloads successfully.	File downloaded successfully.	Pass
TC-03	Check the downloaded filename	File is named students.csv and has a .csv extension.	Filename and extension were correct.	Pass
TC-04	Check CSV column headers	All six required column headers are present.	Required headers were present.	Pass
TC-05	Check exported student details	Student values match the records available in the system.	Exported data matched the available records.	Pass
TC-06	Export multiple student records	Each student is included in a separate CSV row.	Multiple records were exported correctly.	Pass
TC-07	Check values containing special CSV characters	Values remain correctly formatted in the CSV.	CSV values were formatted correctly.	Pass
TC-08	Export with no available student records	Export completes without a runtime error.	No runtime error occurred.	Pass
TC-09	Check access to student records	Existing authentication and administrator authorization remain enforced.	Protected access worked as expected.	Pass
TC-10	Check existing student operations	Add, edit, and delete operations continue to work.	Existing operations worked as expected.	Pass
TC-11	Confirm export does not change student data	Student records remain unchanged after export.	Records remained unchanged.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	**11**
Passed	**11**
Failed	**0**
Pass Rate	**100%**

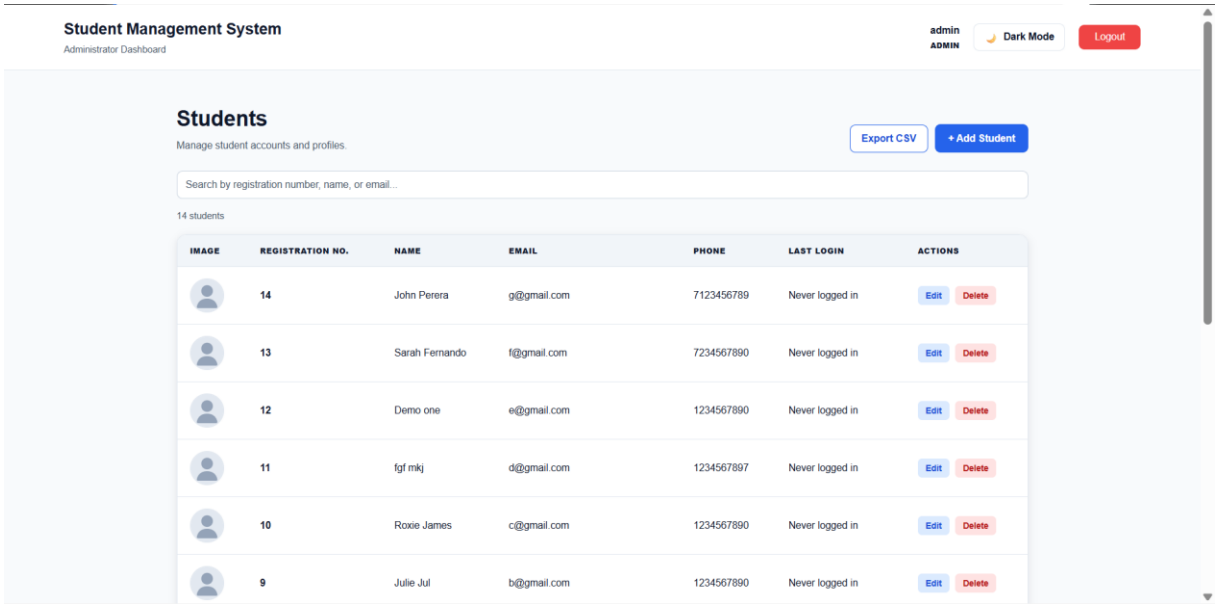
Overall Test Result: Pass

4. Acceptance Criteria Verification

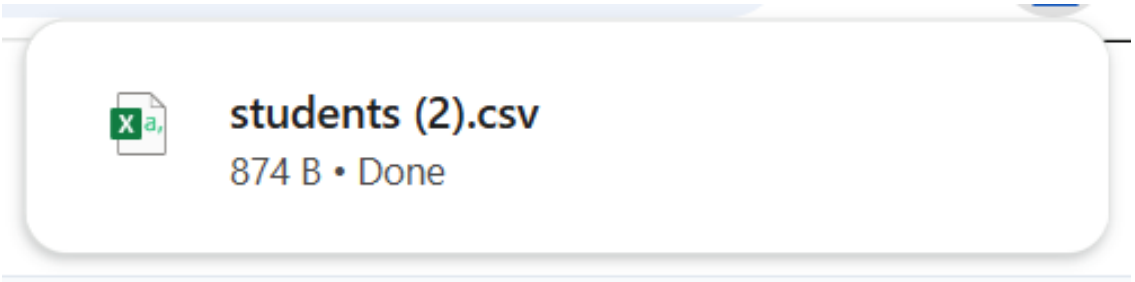
ID	Acceptance Criterion	Status
AC-001	Authorized administrators can access the Export CSV control.	Met
AC-002	Student records can be downloaded in CSV format.	Met
AC-003	The CSV contains the required headers.	Met
AC-004	The CSV contains the required student fields and values.	Met
AC-005	Multiple student records are exported correctly.	Met
AC-006	The file uses the .csv extension and a meaningful filename.	Met
AC-007	Exported data matches the available student records.	Met
AC-008	An empty student list is handled without a runtime error.	Met
AC-009	Unauthorized users cannot access the protected student records.	Met
AC-010	Existing student CRUD functionality continues to work.	Met
AC-011	Existing authentication and authorization remain unchanged.	Met
AC-012	Exporting does not modify or delete student records.	Met

5. Evidence

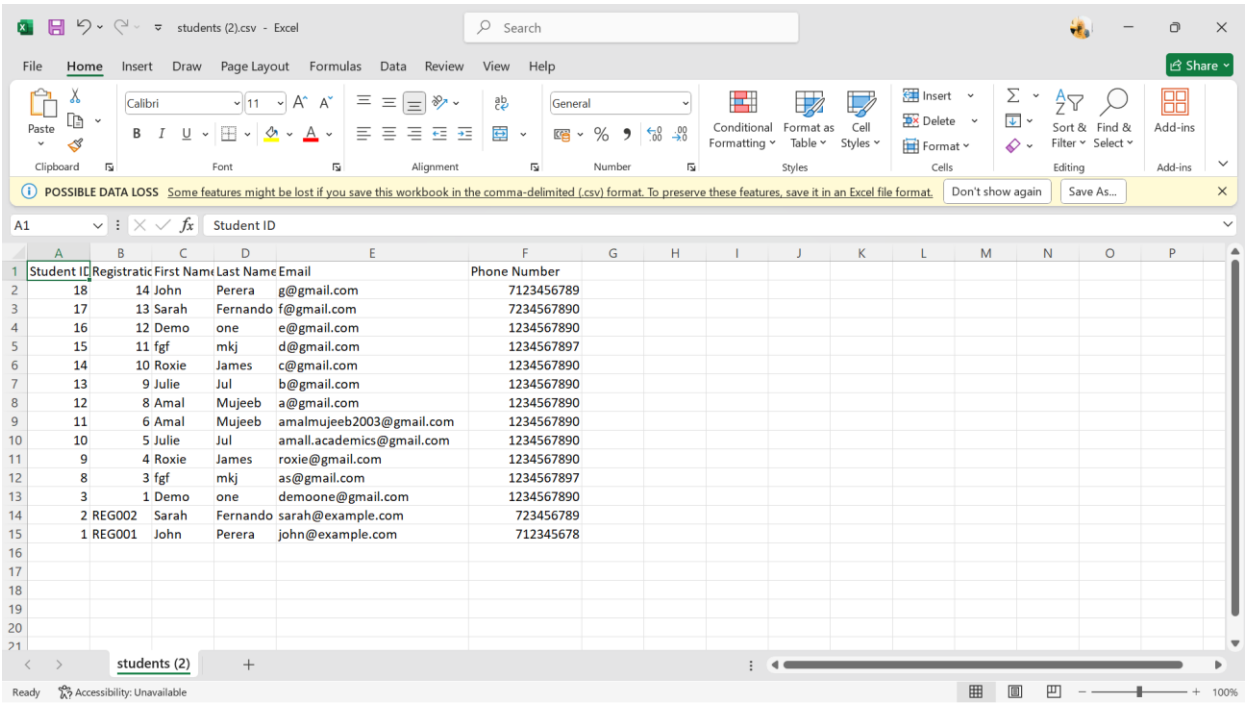
Evidence 01 – Administrator student management page showing the Export CSV control.



Evidence 02 – Successful download of students.csv.



Evidence 03 – Opened CSV file showing the column headers and exported student records.



Student ID	Registratic	First Nam	Last Name	Email	Phone Number
18	14	John	Perera	g@gmail.com	7123456789
17	13	Sarah	Fernando	f@gmail.com	7234567890
16	12	Demo	one	e@gmail.com	1234567890
15	11	fgf	mkj	d@gmail.com	1234567897
14	10	Roxie	James	c@gmail.com	1234567890
13	9	Julie	Jul	b@gmail.com	1234567890
12	8	Amal	Mujeeb	a@gmail.com	1234567890
11	6	Amal	Mujeeb	amalmujeeb2003@gmail.com	1234567890
10	5	Julie	Jul	amall.academics@gmail.com	1234567890
9	4	Roxie	James	roxie@gmail.com	1234567890
8	3	fgf	mkj	as@gmail.com	1234567897
3	1	Demo	one	demoone@gmail.com	1234567890
2	REG002	Sarah	Fernando	sarah@example.com	723456789
1	REG001	John	Perera	john@example.com	712345678

6. Final Result

Item	Result
Requirements	Verified
Implementation	Implemented
Testing	Verified
Acceptance Criteria	Verified
Existing CRUD	Verified
Final Status	Verified

6.1 Final Summary

CR-015 implemented a CSV export facility on the administrator student management page. The exported file contains the required student information, uses appropriate CSV formatting, and downloads as students.csv. The feature reuses the existing student records API and its administrator access controls. No database changes or new backend endpoint were required. Testing was reported as passed, and the existing student management operations remain functional.

6.2 Issues / Limitations

No significant issues were identified during implementation and testing.

Student Name: M M Amal

Student ID: JF/ICT/24/15

Date: 15 October 2026

Signature: Amal